LESSON 2.3: Business Process Design Issues

What this lesson is about

This lesson introduces concepts, design methods, and performance issues encountered during Business Processes creation and implementation. The lesson begins with a review of basic business process concepts and management tasks, including the Graphical Process Modeler (GPM). The processes of running and tracking the results of a business process are covered. It covers design guidelines for business process models and describes modeling strategies for efficient processing.

What you should be able to do

After completing this lesson, you should be able to:

- Describe basic business process concepts.
- Monitor and manage business processes.
- Set performance-related options on the check-in screen.
- Manually run a business process and track results.
- List guidelines for business process design and efficient processing.
- Describe the thread monitor function.
- Explain the guidelines for large file handling.

How you will check your progress

- The progress of the lesson is analyzed based on the successful application of the topics in the scenario and test the exercise independently.

References

Documentation

- IBM Sterling B2B Integrator and BPML

  https://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/com.ibm.help.bpml.doc/SI_SterlingB2BIntegratorAndBPML.html

Business Process management activities

Business Process models

Single business process models generally represent only a portion of an entire business process solution. Typically, users configure multiple business process models to work together to accomplish each business goal. Sterling B2B Integrator is equipped to manage complex operations. Following good design and maintenance practices helps in ensuring that it is optimized for best performance.

The business process models are likely to have decision points and interrelated subprocesses, creating business process models can be complicated by having to manage many business processes at the same time.

The business process shares the aspects of other business processes, such as access to the same software applications, use of the same B2B processing standards, and resources in Sterling B2B Integrator.

Services and Adapters

Flexibility is key to the business analysts who are working with the IT staff to determine how to write business processes. Sterling B2B Integrator enables to write subprocesses with multiple reusable components or as one large service that does a complete process on its own. Fine-tuning these files is important to Sterling B2B Integrator running smoothly.

One way to look at services is to determine where they get the information to process and where they pass it to after it is complete:

- Internal services are located inside Sterling B2B Integrator; they accept parameters and produce results, but they do not interact with any system outside Sterling B2B Integrator.
- Input services receive data from an outside system - adapters, email, AS2.
- Output services send data to an outside system.
- Human interaction services receive or display data through web forms.

Note

Input and output services are known collectively as adapters. Adapters interact with the “world” outside Sterling B2B Integrator. Adapters are special in many ways. For example, adapters can need to be aware of the Perimeter Server (PS), so that they can use the PS to provide communications through a DMZ. Likewise, some adapters (like the MQ adapter and the FTP Server adapter) run in the background and listen for connections. Thus, they are running outside of any business process and do not execute as a step nor generate activity records.

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Business Process management activities

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Business Process files

Files containing business process instructions do not have naming convention. The Sterling B2B Integrator default file name extension is *.bp.

The two separate files that are extracted from a *.bp file:

- *.bpml: The source code for the business process
- *.gpl: The code that tells the GPM how to graphically represent the business process

Business Process development and maintenance

Introduction

Developing and maintaining business process models involves the following tasks:

- Analysis and Design
  - Planning the most efficient processes to achieve the business objectives
  - Determining the Sterling B2B Integrator components that are needed to complete the tasks in the processes
- Coding and Testing
  - Configuring business process model components
  - Creating business process models
  - Testing process models
  - Implementing the business process models in Sterling B2B Integrator
- Maintenance
  - Modifying business process models as changing needs require them to grow, shrink, change, merge, and split
  - Keeping records of the process models you create, use, and modify, for reference

Business Process models as a resource

The business process models using Sterling B2B Integrator resources, and sometimes known as resources or "assets."

You can save them to an offline directory and import and export the files as needed from one copy of Sterling B2B Integrator to another. After the models are created and labeled as to what they were written for, they can be the templates for future process models.

The BP management goal is to maximize the usefulness and efficiency of the BP models that are created.

To minimize set-up time, Sterling B2B Integrator includes a number of predefined BP models that you can use. In addition, follow Sterling B2B Integrator guidelines for BP creation so that the BPs run at the maximum system performance possible for your activities.

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Business Process development and maintenance

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Business Process management tasks

Business processes are managed with the Sterling B2B Integrator environment by enables to test common steps of each business process such as:

- Create a business process
- Validate a business process
- Check In a business process
- Start and stop a business process
- Archive a business process
- Restore a business process
- Delete a business process
- Monitoring a business process

Business Process Implementation

Introduction

After analyzing the business needs, determine the business processes, and document the results. You can then implement and run the business process within Sterling B2B Integrator.

Implementing a business process through GPM

To implement a business process by using the GPM, complete the following steps appropriate to the system requirements:

1. Create a business process model, possibly including:
   - Additional parameters
   - Sub flows
   - Rules and conditions
   - name-value pairs that represent data to locate in process data
   - Correlation service to generate correlations for business processes and document exchange.
2. Validate and save the business process model document.
3. Add the business process to Sterling B2B Integrator (check it in).

Validate a business process

A business process must be validated to run successfully. Therefore, you must ensure that you validate the BPML code in each business process before you add the process to Sterling B2B Integrator.

The Sterling B2B Integrator stores business process definitions in XML, a business process analyst can define them using the GPM or a text editor.

You must validate and compile the process before you can run it.

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Business Process Implementation

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Check in a business process

Before you can use a business process within Sterling B2B Integrator, you must first check it into the system. You can check in a business process model from within the GPM interface or through the Sterling B2B Integrator interface. Each time that you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

To check in a business process by using the Sterling B2B Integrator UI, select Business Process > Manager. In the Create section, next to Process Definition, click Go!.

When you check in a business process, you complete a series of dialog boxes before the business process is considered checked in.

The following sections on the Editor page pertain to system performance:

- Process Levels
- Deadline Setting
- Life Span
- Confirm

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Business Process Implementation

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Check in a business process

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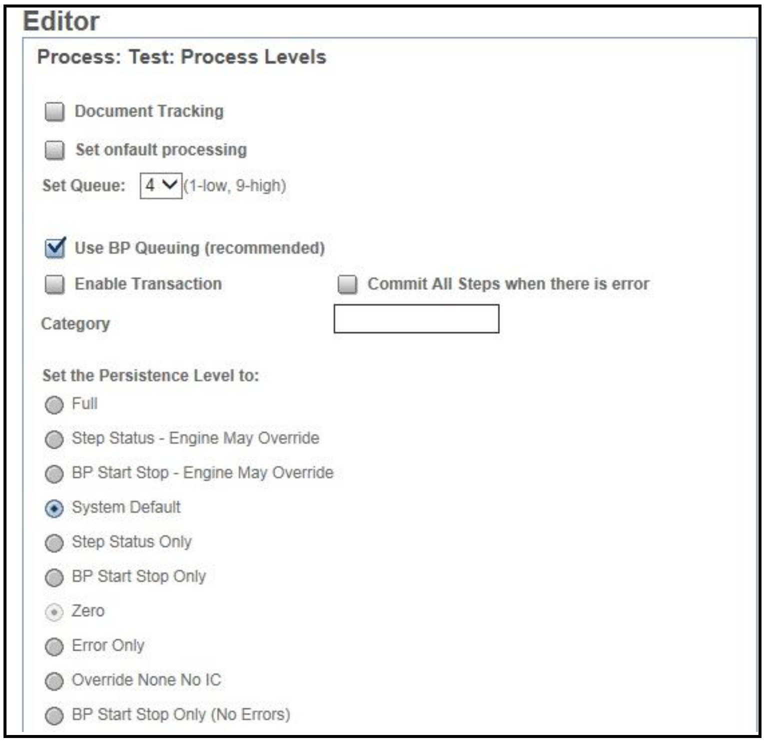
Process level

The following figure shows the fields for the top half of the Editor: Process Levels page



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Business Process Implementation

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Check in a business process

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Process level

Field	Description
Document Tracking	Enabling document tracking provides you with the ability to track a document through its entire lifecycle, both inside and outside your firewall. Document tracking at the business process level is disabled by default, and this setting overrides the global document tracking setting. If you are not using Sterling B2B Integrator document tracking features, leave the option disabled, to reserve system resources for other tasks.
Set On-fault processing	On-fault processing allows the process to immediately run the on-fault activity that is specified in the process, even if the process has not yet reached that step in the process. For example, if a process fails at step 3, but the on-fault activity is specified in step 7, if on-fault processing is enabled, the process proceeds to the step 7 on-fault rather than halting at step 3. Select this option to the enable on-fault activity specified in the process to immediately run if the event of a system error.
Set Queue	You can set performance optimizations by queue in Sterling B2B Integrator. Each Sterling B2B Integrator queue has different levels of allocate resources. Sterling B2B Integrator provides queues Q1 through Q9 with varying levels of resources. Queue Q4 is the default and even system processes are run from there unless they are moved to another queue. Select the queue that is best suited for processing this business process. The default mix of Fairshare queues is tuned and grouped into five main categories as follows: ■ Batch - Q1-3 (trickled, long, and short) ■ Ad Hoc - Q4-5 (Q4 is the default for all business processes) ■ Interactive - Q6-7 (long Cycle, short cycle) ■ Real Time - Q8 ■ System Processes - Q9
Use BP Queuing (recommended)	If this field is selected, then the business process is placed in a queue for processing. This method is the default method. To run the business process in non-queued mode, clear this check box. In non-queued mode, by default the persistence level is set to zero, so the business process cannot be manually restarted or resumed.

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Business Process Implementation

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Check in a business process

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Process level

Field	Description
Enable Transaction	Select this option to instruct Sterling B2B Integrator to treat the entire process as a single transaction so that either all of the steps complete, or, none of them do. When an error occurs, no data is committed; data returns to its pre-process state. By default, this transaction mode is not enabled. Enabling transactions applies only to services that support transaction mode. If transaction management is already built into the process model (the model includes Start and End Transaction services), do not select this option or the process fails. Commit All Steps when there is error: This option prevents partial loads to the database and duplicate information. Select this option to commit all the data to the database even when there is an error in a step or service. When the option is clear, no data is committed to the database if an error occurs.
Commit All Steps when there is an error	When a business process runs with the Enable Transaction option set to true, and an error occurs, all steps up to the error step are committed, if this field is selected. If this field is not selected, then all the steps up to the error steps are rolled back except for the first step and the error step. If all the operations in the error step are transactional, then they are also rolled back.
Category	You can optionally enter a category name to which this process model belongs, creating the category. The category does not affect processing; categories are related to future enhancements.

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Business Process Implementation

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Check in a business process

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Persistence level

Sterling B2B Integrator persist information about nearly every step in a business process. However, reducing persistence comes at a price. The ability of a process to recover in the event of system failure depends on data that is being persisted.

If too low a level of persistence is chosen, it can not be possible to recover correctly – or at all. You cannot turn down persistence without thinking of the repercussions.

But getting persistence right is critical to performance. Incorporate the following suggestions for persistence levels when tuning your system.

For more details, refer:

http://public.dhe.ibm.com/software/commerce/doc/sb2bi/v5r2/SI52_BP_Modeling_book.pdf

Guidelines

- Do not use Full persistence unless your business needs it, as this requires more system resources. The more business process steps and data that is written, the more processor, memory, and database capacity (disk space) are used.
- Use the persistence level of None if your business process does not require data to be accessed further in the process, and you DO NOT need detailed tracking information.
- Use Full persistence when you are testing your processing, to keep more information for troubleshooting scenarios.

Example

If you have an outbound EDI business process that has an X12 Envelope activity, the business process data is translated into EDI and the envelope is applied to the message. When the envelope is applied, Sterling B2B Integrator creates a control number for the document. This control number is unique for each document. If you run this business process several times, each document receives a different control number, so you want to persist the enveloping activity in the business process.

- If the information called by business processes is static (never changing) in the database, you do not need to persist the select Info step in a business process. Instead, you can call the same information from the database at any time.
- For production systems, it is recommended to set persistence low at the global level and set it higher, when required, at the business process or business process step level.

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Business Process Implementation

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Check in a business process

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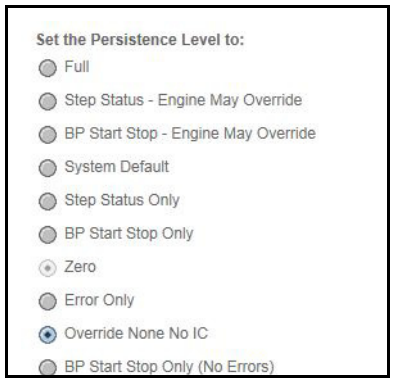
Persistence level

The screen capture and table list describes the fields for the Persistence levels that are shown in the screen capture: Process Levels page. For more details, refer the description in the table:



Persistence level	Description
Full	Retains all data for this BP. This selection uses the most system resources of any persistence level setting.
Step Status - Engine may override	Persists all step data and process data and documents for service steps according to the persistence level supported by the service in the step.
BP Start Stop - Engine May Override	Persists start, stop, and error step data and process data and documents for service steps according to the persistence level supported by the service in the step.
System Default	Persists data according to the global persistence values configured in the noapp.properties file.
Step Status Only	Persists only the status of each step but no process data or documents.
BP Start Stop Only	Persists status information for only the start and stop steps.
Zero	This level does not persist any business process data for recovery or process tracking.
Error Only	Persists information only for error steps.
Override None no IC	The business process encounters an error and stops, the error step is not persisted and the business process stays in the ACTIVE state.
BP Start Stop only no Errors	Does not retain details while the business process is running or in the current process, and will not retain any details after the business process is run.

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Business Process Implementation

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Check in a business process

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Event reporting level

Events are actions that occur within the system. For example, start and stop adapters, business process status or state changes, and errors and exceptions, that are configured to generate event data. Sterling B2B Integrator uses events for various features, such as event-driven notifications.

Note

Full event reporting ensure that all of your system features operate correctly. Minimal or no event reporting settings are options that can improve system performance in some specialized circumstances. However, you use these options only upon recommendation from IBM Customer Support, when Support determines that features dependent on events are not needed.

The following screen capture shows the options for setting the report level:

The following table lists and describes the level options for event reporting:

Field	Description
Full (default)	Generate events for the business process, including the business process start and end time, start and end times for all services or adapters that are running as a result of these business processes, and any resulting errors and exceptions. This is the default setting and selection.
Minimal	Generate events for the business process, including the BP start and end time and any resulting errors and exceptions.
None	Do not generate any event reporting.

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Business Process Implementation

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Check in a business process

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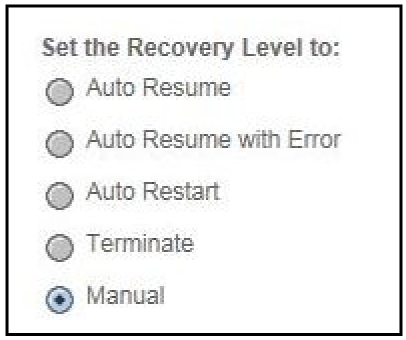
Recovery level

You can set the recovery level for when the business process is interrupted during execution.

The following screen capture shows the options for setting the recovery level:



The following table lists and describes the levels of recovery options:

Recovery level	Description
Auto Resume	Resumes the business process at the last point at which the business process was persisted.
Auto Resume with Error	Goes to the next step as if the step that was interrupted was an error. This allows to roll-back if a non-transactional service is interrupted.
Auto Restart	Terminates the current business process and restarts it from the beginning.
Terminate	Terminates the business process.
Manual (default)	Requires you to resume or restart the business process manually.

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Business Process Implementation

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Check in a business process

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Softstop recovery level

The following screen capture and table list describes the levels of softstop recovery options:

Recovery level	Description
Auto Resume	Resumes the business process at the last point at which the business process was persisted.
Auto Resume with Error	Goes to the next step as if the step that was interrupted was an error. This allows to roll-back if a non-transactional service is interrupted.
Auto Restart	Terminates the current business process and restarts it from the beginning.
Terminate	Terminates the business process.
Manual (default)	Requires you to resume or restart the business process manually.

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Business Process Implementation

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Check in a business process

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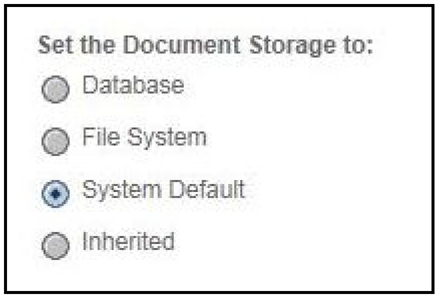
Document storage

You can set the recovery level for when the business process is interrupted during execution. The following screen capture and table lists describes the levels of recovery options:



Storage location	Description
Database	Store documents in the database. Use this option for small documents or when clustering without a clustered file system.
File system	Store documents in the file system. This option can be best for large documents (over 2 G). With this method, archiving and purging the documents from the file system can require special handling.
System Default (default)	Use the defaultDocumentStorageType = DB or FS option that is configured in the jdbc.properties file (either database or file system). As shipped, Database is the document storage type; it can be modified for your installation to use the file system.
Inherited	If this is a subprocess, use the same storage type that is indicated for the parent business process. When one document is created as a result of another, the new document uses the storage type of previous document.

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Business Process Implementation

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Check in a business process

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Deadline settings

A Deadline Setting allows to specify a time for a business process (that is not in a state of ACTIVE) to move to the front of the queued business processes waiting to run.

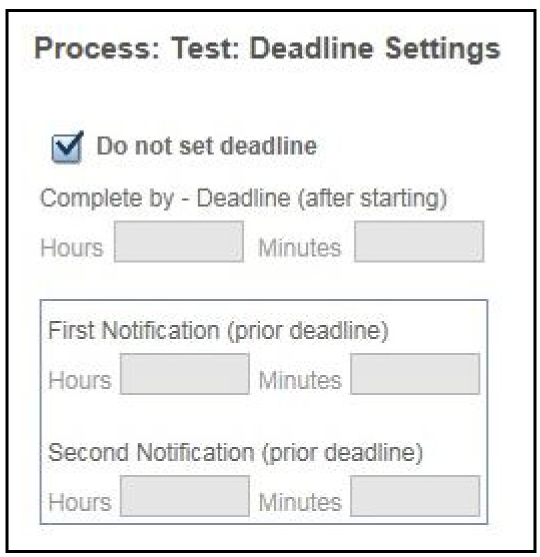
This notifies the system to place the business process in queue to move to the front of the queue. Every effort is made to run the business process at the front of the queue. If the business process in queue misses its queue and reaches the first and second deadline, an exceptional event is fired and posted to the dashboard.

In order for you to receive the notification, you need to define the event posted to dashboard by setting EnableDeadlines=true in noapp.properties for each queue and define the event rule in the eventrule.properties file.

```
EnableDeadlines.1=true for queue 1.
```

To set the Deadline Setting, set it for the number of hours or minutes from the business process start time. By default deadline and notifications are not set. To modify a Deadline Setting, you must modify the business process by checking out the business process, then check it back in.

The following screen capture depicts the fields for the deadline settings:



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Business Process Implementation

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Check in a business process

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Deadline settings

The following table lists describes the fields for the deadline settings:

Field	Description
Do not set deadline (default)	Enables you to run the business process without a deadline.
Complete by - Deadline (after starting)	Enables you to specify a deadline time, in hours and minutes, by which the business process must complete process. If the deadline is reached before the process completes, the system generates a deadline event as the process continues to run.
First Notification (prior deadline)	Hours and Minutes: Enables you to configure sending of a notification to the administrator at a specified number of hours and minutes after a business process starts, if the process is not complete.
Second Notification (prior deadline)	Hours and Minutes: Enables you to specify whether to receive another notification before a business process deadline.

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Business Process Implementation

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Check in a business process

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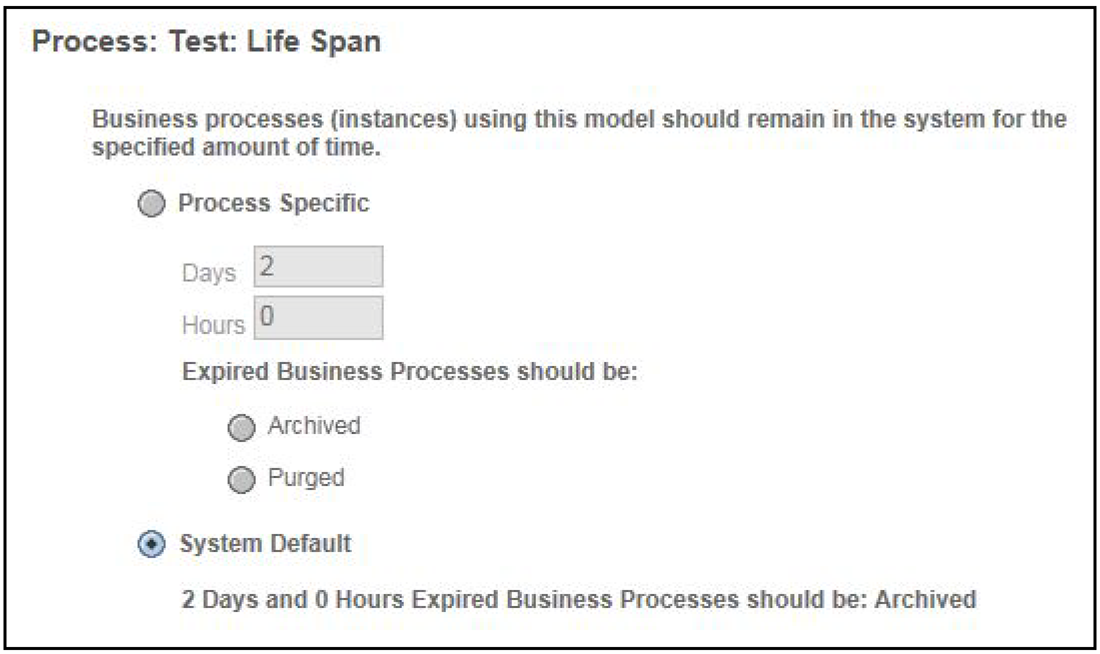
Life span

You can modify the length of time that data pertaining to each instance of this business process model will remain in the system after the process has completed, to be available for monitoring, tracking, and reporting activities, and specify whether to archive or purge the process-related data when the life span expires.

The following screen capture depicts the fields for the life span settings:



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Business Process Implementation

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Check in a business process

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Life span

The following table lists describes the fields for the life span settings:

Option	Description
Process Specific	Enables you to define the length of time to retain business processes and associated data in the system after the process is complete. This allows you to review the transaction data within the retention period. Once a process lifespan is past, then it can be removed from the system by either deleting (purge) or archiving. Archiving allows deleted data to be restored so that the transaction data can be examined. It is not intended for long-term business audit purposes. The default is to keep data for two days and then archive it. Archiving is a fairly expensive operation. Another option is to increase the life span but set the removal method to purge. This keeps the data around longer, in case it is needed, but reduces database space and processing requirements over time.
System Default	For the data, archiving configuration is already defined in Sterling B2B Integrator. The default is two days and the process is archived.

Important

Sterling B2B Integrator Archive Manager, under the Operations menu, is used to change default archive settings archive business processes, view the archive log and restore archived business processes.

Note

Each time that you check in a business process, all the checked in versions of business process is retained in the process model. When you recheck in, the Set Default Version page appears after the Life Span page. It lists all the versions of business process, where you can set any one of the versions as default version.

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Business Process Implementation

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Check in a business process

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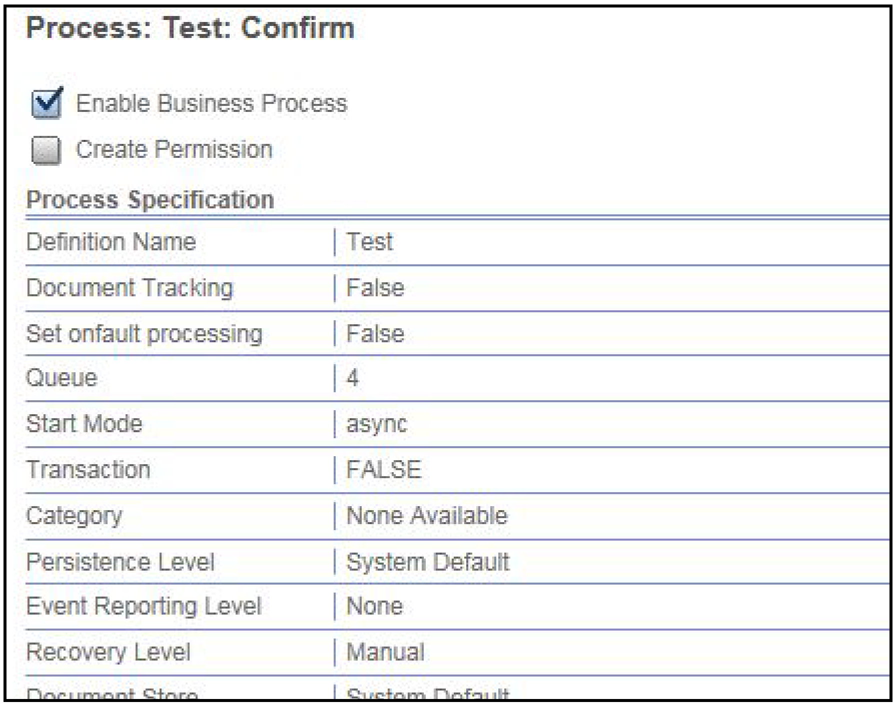
Confirm

You are able to modify your business process model through permissions.

The following screen capture depicts the confirmation page settings:



The following table list describes the confirm page settings:

Option	Description
Enable Business Process	Makes the process model available to run. The business process model is enabled by default. If you are not ready to run the process, clear the check box.
Create Permission	Select this option to have a permission selection that is created for this business process model automatically. The permission can be assigned to user accounts to enable or restrict user access to running the process. Permissions are optional. If none is assigned, the process is available for all users

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Business Process Implementation

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Start a business process

Sterling B2B Integrator can start business processes in different ways:

- According to a user-defined schedule.
- In response to an activity (for example, placement of file into specific folder)
- A user manually starts a process
  - Run from the GUI - Business Process Manager
  - Run from the command line - workflowLauncher.sh

Sterling B2B Integrator supports bootstrapping. You can configure a single input adapter in Sterling B2B Integrator to work outside of a business process and to dynamically select the appropriate business process to run when data enters. Input data must enter Sterling B2B Integrator by way of an input adapter. This adapter takes the received data, puts the data and any metadata it finds into an Initial Business Process Context (IBPC), and begins bootstrapping. A business process definition is selected based on the data and metadata and instantiated as needed.

Run from the GUI - Business Process Manager

Sterling B2B Integrator runs most business processes automatically. However, there are times, when it is necessary to run a business process manually, as when testing. The following steps describe the process to run a business process manually. It assumes that the business process has been validated and checked in.

1. Click Business Process > Manager - The Business Process Manager screen is shown.
2. Type the process name in the Search text box and click Go!.
3. Click the execution manager icon for the process.
4. Click the execute icon.
5. Click Browse and go to the test data file at C:\6F89G\Labfiles\InventoryOf11.xml.
6. Locate the required file and click Open > Go!.
7. Observe the separate Execute Business Process window displayed while the business process is running.
8. Close the Execute Business Process window when the business process is completed.

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Business Process Implementation

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Start a business process

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Run from the command line - workflowLauncher.sh

Sterling B2B Integrator also provides a utility that is called workflowLauncher that allows you to start a business process from the command line. The utility launches a business process and provides the status when it is complete, or when a timeout occurs (whichever comes first).

To launch a business process from the command line:

1. Change to the install_dir/bin directory.
2. Run one of the following commands:
   - ./workflowLauncher.sh -n BPname -f inputFile (UNIX)
   - workflowLauncher.cmd -n BPname -f inputFile (Windows)

Number of options that can be specified. For Example, BPname and inputFile are provided. To view the help screen with the list of options, use the -h or -? Option.

Example

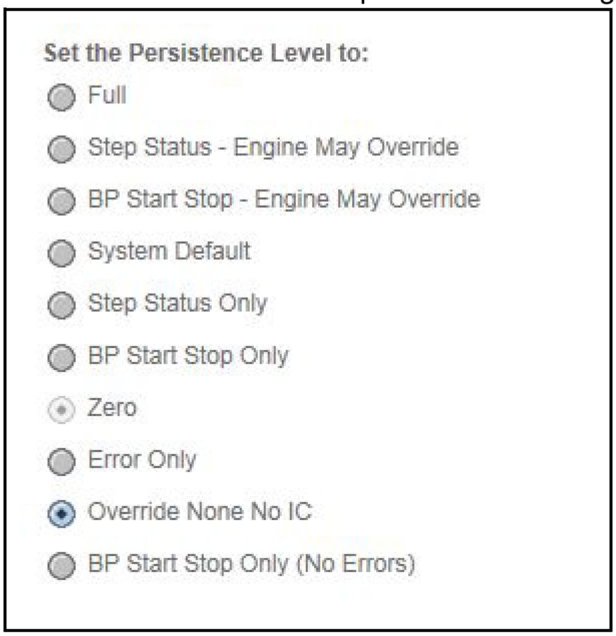
```
./workflowLauncher.sh -n basicinventoryprocess -f C:\6F89G\Labfiles\InventoryOf3.xml
```

Exercise 2.3.1: Persistence level business process

Introduction

Angela checks basicinventoryprocess business process in ten different way to check the different results of changing the persistence levels as per the guidelines provided by an administrator.

In this exercise, Angela learns to check in business process in following ways individually:



Instructions

Full persistence

Complete the following steps to check in the basicinventoryprocess business process with a Full persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

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Exercise 2.3.1:Persistence level business process

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Instructions

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Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_Full** as **Name**.

Step 7: Select Check in Business Process that is created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **Full persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **Full** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

Step 15: In the Search text box, type the **basicinventoryprocess_Full** and click **Go!**.

Step 16: Click the **execution manager** icon for the process.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file.

Step 19: Observe the separate Execute Business Process window that is displayed while the business process is running.

Step 20: Close the Execute Business Process window when the business process is complete.

Step 21: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of Full option.

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Exercise 2.3.1:Persistence level business process

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Instructions

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Step Status - Engine May Override

Complete the following steps to check in the basicinventoryprocess business process with a Step Status - Engine May Override persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_stepstatusengineoveride** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **Step Status - Engine May Override persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **Step Status - Engine May Override** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 15: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 16: Click the **execution manager** icon for the process you checked in.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file. Observe the separate Execute Business Process window that is displayed while the business process is running.

Step 19: Close the Execute Business Process window when the business process is complete.

Step 20: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of Step Status - Engine May Override option.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

BP Start Stop - Engine May Override

Complete the following steps to check in the basicinventoryprocess business process with a BP Start Stop - Engine May Override persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_BPstartstopengineoveride** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **BP Start Stop - Engine May Override persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **BP Start Stop - Engine May Override** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 15: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 16: Click the **execution manager** icon for the process you checked in.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file. Observe the separate Execute Business Process window displayed while the business process is running.

Step 19: Close the Execute Business Process window when the business process is complete.

Step 20: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of BP Start Stop - Engine May Override option.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

System Default

Complete the following steps to check in the basicinventoryprocess business process with a System Default persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_System Default** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **System Default persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **System Default** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 15: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 16: Click the **execution manager** icon for the process you checked in.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file. Observe the separate Execute Business Process window displayed while the business process is running.

Step 19: Close the Execute Business Process window when the business process is complete.

Step 20: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of System Default option.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step Status Only

Complete the following steps to check in the basicinventoryprocess business process with a Step Status Only persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_StepStatusOnly** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **Step Status Only persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **Step Status Only** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 15: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 16: Click the **execution manager** icon for the process you checked in.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file. Observe the separate Execute Business Process window displayed while the business process is running.

Step 19: Close the Execute Business Process window when the business process is complete.

Step 20: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of Step Status only option.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

BP Start Stop Only

Complete the following steps to check in the basicinventoryprocess business process with a Start Stop Only persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_BPstartstoponly** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **BP Start Stop only persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **BP Start Stop Only** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 15: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 16: Click the **execution manager** icon for the process you checked in.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file. Observe the separate Execute Business Process window displayed while the business process is running.

Step 19: Close the Execute Business Process window when the business process is complete.

Step 20: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of BP Start Stop only option.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Error Only

Complete the following steps to check in the basicinventoryprocess business process with a Error only persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_Erroronly** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **Error only persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **Error Only** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 15: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 16: Click the **execution manager** icon for the process you checked in.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **Test.txt** file. Observe the separate Execute Business Process window displayed while the business process is running.

Step 19: Close the Execute Business Process window when the business process is complete.

Step 20: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of Error only option.

Note

Observe the result to view the error steps, if you upload InventoryOf3.xml file steps is not displayed.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Override None No IC

Complete the following steps to check in the basicinventoryprocess business process with a Override None No IC persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_OverrideIC** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **Override None No IC persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **Override None No IC** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 15: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 16: Click the **execution manager** icon for the process you checked in.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file. Observe the separate Execute Business Process window displayed while the business process is running.

Step 19: Close the Execute Business Process window when the business process is complete.

Step 20: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of Override None No. IC option.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

BP Start Stop Only (No Errors)

Complete the following steps to check in the basicinventoryprocess business process with a BP Start Stop Only (No Errros) persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_BPstartstopnoerrors** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Enter **BP Start Stop Only no errors persistence** for the description and click **Next** to continue.

Step 10: Set the persistence level to **BP Start Stop Only (No Errors)** and accept the remaining defaults. Click **Next** to continue.

Step 11: Do not set a deadline for this process, click **Next** to continue.

Step 12: Accept the default for life span and click **Next** to continue.

Step 13: Verify that the business process is enabled and click **Finish**.

Step 14: Click **Return**. The Business Process Manager screen is shown.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 15: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 16: Click the **execution manager** icon for the process you checked in.

Step 17: Click the **execute** icon.

Step 18: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file. Observe the separate Execute Business Process window displayed while the business process is running.

Step 19: Close the Execute Business Process window when the business process is complete.

Step 20: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of BP Start Stop Only (No Errors) option.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Zero persistence

Complete the following steps to check in the basicinventoryprocess business process with a Zero persistence level:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_zero** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file name.

Step 9: Check **Enable Transaction**.

(Continued on next page)

Exercise 2.3.1:Persistence level business process

(Continued)

Instructions

....(Continued)

Step 10: Enter **Zero persistence** for the description and click **Next** to continue.

Step 11: Set the persistence level to **Zero** and accept the remaining defaults. Click **Next** to continue.

Step 12: Do not set a deadline for this process, click **Next** to continue.

Step 13: Accept the default for life span and click **Next** to continue.

Step 14: Verify that the business process is enabled and click **Finish**.

Step 15: Click **Return**. The Business Process Manager screen is shown.

Step 16: In the Search text box, type the **basicinventoryprocess** and click **Go!**.

Step 17: Click the **execution manager** icon for the process you checked in.

Step 18: Click the **execute** icon.

Step 19: In the Filename dialog box, browse to **C:\6F89G\Labfiles** folder and select **InventoryOf11.xml** file. Observe the separate Execute Business Process window displayed while the business process is running.

Step 20: Close the Execute Business Process window when the business process is complete.

Step 21: Access **Business Process** > **Monitor** > **Current Process** to view the step wise result of persistence level of Zero option.

Note

For more details, refer from this link.
http://public.dhe.ibm.com/software/commerce/doc/sb2bi/v5r2/SI52_BP_Modeling_book.pdf

(Continued on next page)

Exercise 2.3.2: Document Storage Settings

Introduction

Angela needs to check in a business process with the different document storage type options available in a business process to check in and view the results for each option.

The different Document Storage setting available are:

- Database
- File System
- System Default
- Inherited

Instructions

Database

Complete the following steps to check in a business process with the document storage type as Database:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_DatabaseStorage** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file

Step 9: Enter **Database Storage** for the description and click **Next** to continue.

Step 10: Set the persistence level to **Full.**

Step 11: Set the document storage type as **Database** and accept the remaining defaults. Click **Next** to continue.

Step 12: Do not set a deadline for this process, click **Next** to continue.

(Continued on next page)

Exercise 2.3.2:Document Storage Settings

(Continued)

Instructions

....(Continued)

Step 13: Accept the default for life span and click **Next** to continue.

Step 14: Verify that the business process is enabled and click **Finish**.

Step 15: Click **Return**. The Business Process Manager screen is shown.

Step 16: In the Search text box, type the **Database** and click **Go!**.

Step 17: Click the **execution manager** icon for the process.

Step 18: Click the **execute** icon.

Step 19: In the Filename dialog box, browse to and select **InventoryOf11.xml** from your Lab Files folder. Observe the separate Execute Business Process window displayed while the business process is running.

Step 20: Close the Execute Business Process window when the business process is complete.

Note

You can view the latest entries of the executed business process in the database. The documents that are related to the business process are stored in the database based on the chosen document storage type.

Important

The payload of each document in the system is stored in the DATA_TABLE or TRANS_DATA tables. When documents (payloads) become large, it leads to inefficient use of database disk space and results in greatly increased network traffic.

(Continued on next page)

Exercise 2.3.2:Document Storage Settings

(Continued)

Instructions

....(Continued)

File System

Complete the following steps to check in a business process with the document storage type as File System:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Business Processes** > **Manager.** The Business Process Manager page is displayed.

Step 5: Click **Go!** next to **Process Definition** under **Create**.

Step 6: Type **basicinventoryprocess_FileSystemStorage** as **Name**.

Step 7: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 8: Browse to the **C:\6F89G\Labfiles** folder and select **basicinventoryprocess.bp** for the file

Step 9: Enter **File System Storage** for the description and click **Next** to continue.

Step 10: Set the persistence level to **Full.**

Step 11: Set the document storage type as **FileSystem** and accept the remaining defaults. Click **Next** to continue.

Step 12: Do not set a deadline for this process, click **Next** to continue.

(Continued on next page)

Exercise 2.3.2:Document Storage Settings

(Continued)

Instructions

....(Continued)

Step 13: Accept the default for life span and click **Next** to continue.

Step 14: Verify that the business process is enabled and click **Finish**.

Step 15: Click **Return**. The Business Process Manager screen is shown.

Step 16: In the Search text box, type the **FileSystem** and click **Go!**.

Step 17: Click the **execution manager** icon for the process.

Step 18: Click the **execute** icon.

Step 19: In the Filename dialog box, browse to and select **InventoryOf11.xml** from your Lab Files folder. Observe the separate Execute Business Process window displayed while the business process is running.

Step 20: Close the Execute Business Process window when the business process is complete.

Step 21: Click FileZilla from the desktop to access **/opt/IBM/SterlingIntegrator/install/documents** folders to view the file that is created in the folder structure based on the system current date.

Note

You can view the file that is created for the executed business process. The documents that are related to the business process are stored in the default documents folder based on the chosen document storage type.

Important

This option was created to store the payload of documents out on disk. For documents with file system persistence, the DATA_TABLE or TRANS_DATA tables are still used. The difference is that the content of the BLOB data is not the payload, but a serialized Java HashMap that contains the file name of the payload.

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Exercise 2.3.2:Document Storage Settings

(Continued)

Instructions

Manually tuning Default document storage

You can configure the default storage type in the jdbc.properties file. The default storage type as Database.

To view the default storage type settings:

Step 1: Access windows image, from the desktop click FileZilla icon.

Step 2: Log in Sterling B2B Integrator installs directory through FileZilla.

Step 3: Go to the path **/opt/IBM/SterlingIntegrator/install/properties**

Step 4: Browse and open jdbc.properties.

Step 5: Search for defaultDocumentStorageType. You can view the default value set as **DB**. To make file system as default storage type, you can set the defaultDocumentStorageType as **FS.**

Step 6: The default path where the files are created for the File System document storage type is set with the parameter document_dir.

Step 7: Search for document_dir and view the default path setting as **/opt/IBM/SterlingIntegrator/install/documents**. If required you can change the default path with the required path.

Note

On changing any values in jdbc.properites file, you need to stop the Sterling B2B Integrator and run setupfiles.sh and then start the Sterling B2B Integrator to view the changes.

Requirements

On Editing any properties file you need to run setupfiles.sh and restart the Sterling B2B Integrator.

Inherited storage

If a subprocess is configured as Inherited document storage type, then it uses the same storage type that is indicated for the parent business process. When one document is created as a result of another, the new document uses the storage type of previous document.

Monitor business process results

Introduction

You can monitor run business processes by using the Business Monitor page and the Business Process Monitor Advanced Search page.

You can view the record of executed business process instance data by using the business process monitor. To view the record of executed business process instance data:

1. Click **Business Process > Current Processes**. The Business Process Monitor screen shows the current processes.
2. Locate first instance of your process in the list.
3. Click the corresponding ID to view the **Business Process Detail** page.
4. Click the **Info** icon in the **Document** column or the **Instance Data** column of any of the steps to view the detail information for the step.

Business Process Monitor Advanced Search

The Advanced Search function is a powerful tool to locate and analyze previously run business processes. Options are provided to search by: Process ID or BP name. Search criteria can be narrowed by specifying: State, Status, Date range, and Deadline.

Note

If the “Enable Business Process Operations” box is selected, business processes in a particular state can be managed, by using activities such as restart, resume, and terminate.

To access the business process results, locate previously run business processes: From the Business Process menu, select Monitor > Advanced Search > Business Processes

Guidelines for business process creation

Introduction

You can avoid many potential performance problems by implementing well-designed business process models. The practices that are described helps to ensure optimum performance of your business processes in Sterling B2B Integrator. For information-specific related too large file handling, see Modeling Guidelines for Large File Handling later in this lesson.

Note

Generally speaking, strategies with the highest potential impact on performance are listed first.

The following list describes the designing structure and naming business process models:

Name Versions Meaningfully

Sterling B2B Integrator stores each version of a business process model that you check in enables you to revert to use a previous version. Therefore, you employ a strategy for using the description field to differentiate business process models as you check them in to Sterling B2B Integrator, to easily identify a previous version. For example, you might add a version number or identify the change that is made to the process.

Retain Service Names in Graphical Process Models

Although you can change the names of services in your graphical process models to make them more descriptive, a good practice is to retain the service name within new name, for future reference you can easily see which service you used for the step. Changing the name of a service in the GPM changes the display name that is shown in the workspace for that service, but it does not rename the service configuration that is used in the process model.

Retaining the name of the service is helpful so that when you need to update multiple business process models to use a new service (such as when one that you use is retired), you can see the service name in the process model to save research time.

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Guidelines for business process creation

(Continued)

Introduction

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Reusability

Design reusable business processes whenever possible. Create business processes with fewer steps that you can reuse so that you can avoid the work of creating new ones.

- Create common subprocesses that can be called from many parents
- Build a library of common subprocesses
- Create parent processes that can be templates for large processes (calling different child processes)
- Create reusable processes that can be used in sequences of processing, instead of having one parent process
- Control the overall flow of subprocesses, a series of processes are chained together and run in sequence.
- Build parameterized business processes that can configure at run time or business process context

Use Predefined Business Processes

Whenever possible, use predefined process models provided with Sterling B2B Integrator, to save work.

Keep Subprocesses Manageable

Keep each subprocess to a manageable size. If the subprocess runs multiple activities, consider splitting it into multiple, smaller subprocesses. This approach enhances reusability and simplifies maintenance.

Another consideration is how you intend to start subprocesses. If you are using inline invocation of subprocesses, the impact on performance is relatively small, whereas asynchronous starts can have a more marked negative effect on performance. If you need to use asynchronous starts for your subprocesses, consider building the steps into the parent process instead.

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Guidelines for business process creation

(Continued)

Introduction

....(Continued)

Invoke subprocesses asynchronously when appropriate as listed in the following points:

- Start subprocesses asynchronously when running in parallel with the main process makes sense, such as in a circumstance in which you need activities to be completed,.
- Unlike using the BPML All activity, in this case, the activities do not synchronize as they do at the end of the All activity, and changes made to process data in the subprocesses are not available in process data.
- When you run asynchronous processes, the communication backs to the parent process unless you include Produce and Consume activities in your process model. The times when it is necessary to have processes run asynchronously, and the Produce/Consume activities in place for such processes to communicate. But, inter-process communication is very slow (whether using produce/consume or other mechanisms).
- Use Produce and Consume services instead of the Invoke Subprocess service to start subprocesses is a business process design flaw.
- Use Produce and Consume services places a business process into a waiting state, waiting for the Produce service to create the document and for the Consume service to use the document.
- Inline starts over async starts, as you are likely to come away with “async invokes are bad, inline invokes are good.” Blindly replacing async invokes with inline invokes is a recipe for disaster, as they would then lose parallelism.
- Working in multiple threads in an async manner is the only way that high throughput can be achieved. However, calling a business process without the intent of spawn another thread to increase parallelism probably need to be done by using inline invoke. Unless you have a business need for an invoked business process to run as a separate process, use the inline invoke option. Processes that are invoked by using the inline option that is run synchronously as part of the original process, saving the processing time of starting a new unique process or thread.

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Guidelines for business process creation

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Introduction

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Make Modifications Incrementally

Make incremental modifications to your business processes, and thoroughly test each change. This approach can save you a great deal of time because incremental changes are easier to troubleshoot than an entire business process.

Use appropriate components

The most current services or adapters available in Sterling B2B Integrator. Many newer services replace the activities of more than one older adapter or service, to reduce the number of steps in a business process, increasing processing efficiency. You can reduce calls to the database, freeing system resources for other processing. To quickly view the list of retiring adapters, click View > Stencil > Retiring in the GPM.

Deprecation Tool

A deprecation tool shows a report of the services, business processes, and maps that are being retired. The deprecation report is consisted of subreports from each clump that has a retired resource to report. The users are notified of the retired services so they can begin the migration process. Each entry in the sub report has a description about the documents in the migration process to the replacement resource along with the retiring date of the old resource.

Run batch transactions where possible

Batch transactions instead of individual transactions whenever appropriate. Running batch transactions reduces the number of calls to the database and completes more processing with fewer adapter and service invocations. You can use this technique to improve processing times and manage workload, by reducing processing during peak processing times and maximizing processing opportunities during non-peak processing times.

Example

You can have a purchase order business process that sends all final purchase orders to a file directory, and a business process that picks up all of the purchase orders from the file directory for processing. If each purchase order is processed when it is received in the file directory, several calls to the database can be made and many invocations made that use system resources.

Guidelines for business process creation

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Introduction

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Test for execution speed and modify business process models as needed

In creating your process models, address speed of execution last. Identify which subprocesses can run in parallel or asynchronously from other subprocesses, and change them after your tests confirm that the subprocess or business process is working.

Limit Steps in a business process Model

Do not include excess steps or activities in a process model, such as loops and redundant calls to the database for information that is retained in the process data. Keep the number of service and adapter configurations in the process to a minimum. Excess activities require more system resources. Fewer steps mean shorter processing time. It also facilitates reuse.

Loops

A common thing that you see is putting N items into PD, and then doing a loop to process each item. The problem with loop is that the time to execute is on the order of N^2 which means that unless N is small this is going to be incredibly slow. Why N^2? Short answer – the cost of dealing with PD goes up linearly for each item in PD.

Example

N items in PD indicate order N more expensive than one item in PD. This price is paid for every step in the loop – and the loop itself is executed N times – giving a total cost of order N * N = N^2. In some cases, it makes sense to use the translator to do the processing that is going on in the loop.

For certain applications (for example, EDI) services that prevent the need for a loop can be provided as discussed. Lastly, it can be possible to put all the work that is done in the loop into a different process that is invoked from within the loop. If only one item is passed to the child process, then you get rid of one factor of N, making the process go down to order N instead.

Minimize Database Communication

Database communication is relatively slow – in general it takes several milliseconds to get a response to any query. Database can not be slow, but it really adds up if you are dealing with a high volume situation. Ideally database communication gets aggregated into as few steps in the process as possible. You can accomplish by clever use of SQL and the lightweight JDBC service or use the translator to handle it.

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Guidelines for business process creation

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Introduction

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Use Schedules to Manage Workload

- Use scheduling features to manage workloads and improve processing times by moving lower priority processing to off-peak processing times. Schedule your lower-priority business processes to run during non-peak times frees resources during peak processing times, improving system, performance and processing times
- Refine your schedules by excluding busy times and days of the week, or month. For example, you can schedule processes to not run on days you process sales orders in your ERP system.
- Schedule processes to run only as often as necessary to meet your business requirements.

Do not schedule too often. However, be careful to execute often enough to handle business needs, and watch for performance problems. Many times, a business process must loop across all the docs that were retrieved, This can be an issue if many documents to loop over, and running the business process once or twice a day can result in large numbers of documents piling up to be processed.

While this task can be done during a non-peak time, the execution time can be long enough to carry over into a peak time. If the schedule configured for an hour or so, then it can be often enough to prevent the docs from piling up and causing this problem.

Use Wait and Sleep services appropriately

The Sleep service holds onto its processing thread (by using more system resources), while the Wait service does not. Use the Wait service instead of the Sleep service for wait times longer than 1 minute, but see the following information for the applicability of each service to your business needs:

- For short periods, the Sleep service can be more efficient, the system does not need to requeue the process.
- Using a Wait service is likely to make more system resources available.
- Sleep business processes can exhaust resources, resulting in Sterling B2B Integrator being unable to execute more processes. Thus, ensure that few processes are sleeping at any one time, and that sleep periods are limited, is critical.

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Guidelines for business process creation

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Introduction

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- The Sleep service enable to set an exact length of time for the process to sleep. The Wait service is far less precise; depending on your Sterling B2B Integrator configuration, the Wait service can wait an extra 10 minutes. If timing is critical, use the Sleep service.

Note

If a number of business processes are running and by using the sleep service, they can use up many available threads.

Limit calls to the file system

If you need repeated access to data from a document on the file system, adding the data from the document to process data (PD) can be the more efficient. In process data, the information can be referenced throughout the life of the process, if the amount of data is small. In large amounts of data, process data can slow down every step in a business process.

Use BPML All activity sparingly

The BPML activity, All (represented in the GPM by the All Start and All End icons), runs two or more complex child activities within a business process simultaneously. The performance implications to using the All activity, use it only if you expect branches of your business process model to take significant amounts of time (in seconds or minutes) to complete

Do not use the All activity to run two or more instances of the same subprocess simultaneously. The system handles the appropriate load balancing without the need. Using BPML All results in a time savings only if it takes a long time to run the steps you are running in parallel. The only things that take long enough for a BPML All to be worthwhile can involve human interaction. Even translation of large documents runs so fast that it does not make sense to use an all (or it makes more sense to run separate processes).

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Guidelines for business process creation

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Introduction

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Manage Process Data

Large amounts of data in PD can make every step of a process go slower. Usually process data size becomes an issue when you put the entire contents of a document into process data.

Large amount of data of the document contents can be converted and put into XML. And, then do XPath manipulation against that DOM for just that step.

For example, if you wanted to count the number of LineItems in a purchase order you might do something like:

<assign to="count" from="count(DocToDOM(PrimaryDocument)/LineItem)"/>. This setting enable you to do XPath processing on the document without dumping it into process data.

The cost of converting a document to DOM is fairly high as well. The user can also use the Release service to remove data from PD after they are done with it.

Pass child processes only what they need

It might seem handy to rely on the default behavior of a child process getting a complete copy of the process data of the parent. However, the cost can be high. In particular, not only is PD copied for the child, but oftentimes the document content itself is copied – which is expensive.

As a rule, unless PD is trivial and only 1 or 2 small documents that are tied with a process, it passed the child process only what it needs by using message_to_child.

Obtain process information in OnFault layer

If you need only process information if a fault in the process flow (where you have an onFault activity in the process model), add a process information service such as the Get Document Info service or the BP Metadata Info service to the onFault layer (subflow), rather than to the main flow of the business process model.

Ensure that relevant onFaults are present

Processing of onFaults should be aimed are expected and unexpected errors. The goal is to reduce the chances of a business process in Halted/Interrupted state.

Develop service as last resort

If testing shows that a step in a process is a serious performance problem - for example, certain loops - and the available constructs/services for business processes do not solve the problem, consider writing this step as a service. For example, Java code instead of BPML.

Use Specified Path in XPath Searches

Introduction

When making an XPath reference to data, by using a specified path is more efficient than a "//" search, and the potential difference in processing efficiency can be seen. Although using the "//" search works, it requires more system resources to complete the search.

Using a "//" search finds every possible node that matches the selection criteria, regardless of how deeply nested it is. This extensive search requires every element in process data to be examined. If the search is done within a loop, processing can become slower as the size of process data increases.

Note

Write XPath statements by using absolute paths. For example, writePurchaseOrder/text instead of /ProcessData/PurchaseOrder/text().

Complex XPath expression

The XPath expression language is fairly powerful. Unfortunately, XPath evaluation is expensive, particularly with complex predicate expressions. Developers minimize, especially in the critical path of performance sensitive processes.

The Invoke business process service

Introduction

The Invoke Business Process Service is used to start an instance of a subprocess (child) from within the current instance of the business process (parent).

Functions of invoke business process service

The Invoke Business Process Service, when used, instantiates an instance of a subprocess, passing the current workflow context and the primary document to the new instances.

- When the Invoke Business Process Service is set to synchronous mode, the parent suspends processing until it receives data from the child. In synchronous mode, the parent is notified when the child encounters errors.
- When the Invoke Business Process Service is set to asynchronous mode, the parent and child process data simultaneously and independently of each other. Therefore, the parent does not receive notification when the child encounters errors.

Steps to implement invoke business process service

To implement the Invoke Business Process Service, complete the following tasks:

1. Create an Invoke Business Process Service configuration.
2. Create and enable a business process that includes the Invoke Business Process Service.
3. Test the business process and the service.
4. Run the business process.

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The Invoke business process service

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Graphical Process Modeler configuration

The following table describes the fields that are used to configure the Invoke Business Process Service in GPM:

Field	Description
Config	Name of the service configuration.
COPY_SERVICE_PARMS	Invoke Business Process Service parameters passed to the subprocess. Valid values are True (default) and False.
Business Process Definition Name (WFD_NAME)	Business process that is used in the service configuration. Valid values include all the business processes.
Invoke Mode (1=Async, 2=Sync 3=Inline) (INVOKE_MODE)	Mode in which to run the subprocess. Valid modes are Asynchronous (default) and Synchronous.
List of Parms to override (PARM_LIST)	Parameters to override.
Type of Error On Which to Notify Parent Process (NOTIFY_PARENT_ON_ERROR)	Errors the subprocess reports to the parent business process (synchronous mode only). Valid values include: Continue parent only if a service-generated error occurs in the subprocess. Continue parent only if a system-generated error occurs in the subprocess. Continue parent if any type of error occurs in the subprocess. Halt subprocess if any error occurs (for potential correction and resume). This option does not notify the parent of the error from the subprocess, and the subprocess is halted.

Exercise 2.3.3: Monitor invoke mode configuration

Introduction

Angela wants to invoke basic inventory business process through a sub process. In the sub process Angela uses invoke business process service and edits the INVOKE_MODE as an asynchronous, synchronous, and inline mode.

In this exercise, Angela tests the behavior and analyzes the importance of the three types of invoke mode.

Instructions

Complete the following steps to configure an Invoke Business Process Service:

Step 1: From the Window image, open a web browser to access **http://192.168.40.100:9000/dashboard** URL.

Step 2: Log in by specifying **admin** as user ID and **password** as password.

Step 3: Click **Sign In**.

Step 4: Access the **Administration Menu** > **Deployment** > **Services > Configuration**. The **Services Configuration** page opens.

Step 5: Under Create, next to New Service, click **Go!**.

Step 6: Select **Invoke Business Process Service** and click **Next**.

Step 7: Enter the Name as **Invoke_Test**

Step 8: Type the description as **invoke business process**.

Step 9: Click **Next**.

Step 10: Confirm the fields that are specified and click **Finish**.

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Exercise 2.3.3:Monitor invoke mode configuration

(Continued)

Instructions

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Complete the following steps to check in the Inline business process:

Step 1: Select **Business Processes > Manager** from the **Administration** Menu. The Business Process Manager page is displayed.

Step 2: Click **Go!** next to Process Definition under Create.

Step 3: Type **Inlineprocess** as **Name**.

Step 4: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 5: Browse to the **Lab Files** and select **Inline.bp** for the file name.

Step 6: Enter **Test inline invoke mode** for the description and click **Next** to continue.

Step 7: Set the persistence level to be **Full** and accept the remaining defaults, click **Next** to continue.

Step 8: Do not set a deadline for this process, click **Next** to continue.

Step 9: Take the default for life span, click **Next** to continue.

Step 10: Verify that the business process is enabled.

Step 11: Click **Finish**

Step 12: Click **Return**. The Business Process Manager screen is shown.

Step 13: In the Search text box, type the **Inline** and click **Go!**.

Step 14: Click the **execution manager** icon for the process.

Step 15: Click the **execute** icon.

Step 16: In the **Filename** dialog box, browse to C:\6F89G\Labfiles\InventoryOf11.xml. Observe the separate Execute Business Process window displayed while the business process is running.

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Exercise 2.3.3:Monitor invoke mode configuration

(Continued)

Instructions

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Complete the following steps to check in the Synchronous business process:

Step 1: Select **Business Processes** > **Manager** from the **Administration** Menu. The Business Process Manager page is displayed.

Step 2: Click **Go!** next to Process Definition under Create.

Step 3: Type **Syncprocess** as **Name**.

Step 4: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 5: Browse to the **Lab Files** and select **Inline.bp** for the file name.

Step 6: Enter **Test sync invoke mode** for the description and click **Next** to continue.

Step 7: Set the persistence level to be **Full** and accept the remaining defaults, click **Next** to continue.

Step 8: Do not set a deadline for this process, click **Next** to continue.

Step 9: Take the default for life span, click **Next** to continue.

Step 10: Verify that the business process is enabled.

Step 11: Click **Finish**

Step 12: Click **Return**. The Business Process Manager screen is shown.

Step 13: In the Search text box, type the **Sync** and click **Go!**.

Step 14: Click the **execution manager** icon for the process.

Step 15: Click the **execute** icon.

Step 16: In the **Filename** dialog box, browse to C:\6F89G\Labfiles\InventoryOf11.xml. Observe the separate Execute Business Process window displayed while the business process is running.

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Exercise 2.3.3:Monitor invoke mode configuration

(Continued)

Instructions

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Complete the following steps to check in the Asynchronous business process:

Step 1: Select **Business Processes** > **Manager** from the **Administration** Menu. The Business Process Manager page is displayed.

Step 2: Click **Go!** next to Process Definition under Create.

Step 3: Type **Asyncprocess** as **Name**.

Step 4: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 5: Browse to the **Lab Files** and select **Inline.bp** for the file name.

Step 6: Enter **Test Async invoke mode** for the description and click **Next** to continue.

Step 7: Set the persistence level to be **Full** and accept the remaining defaults, click **Next** to continue.

Step 8: Do not set a deadline for this process, click **Next** to continue.

Step 9: Take the default for life span, click **Next** to continue.

Step 10: Verify that the business process is enabled and click **Finish**

Step 11: Click **Return**. The Business Process Manager screen is shown.

Step 12: In the Search text box, type the **Async** and click **Go!**.

Step 13: Click the **execution manager** icon for the process.

Step 14: Click the **execute** icon.

Step 15: In the **Filename** dialog box, browse to C:\6F89G\Labfiles\InventoryOf11.xml. Observe the separate Execute Business Process window displayed while the business process is running.

Message Tags

Introduction

In the Invoke Sub-Process service or Invoke Business Process service, by default all the process data passes from the parent process to its child subprocess.

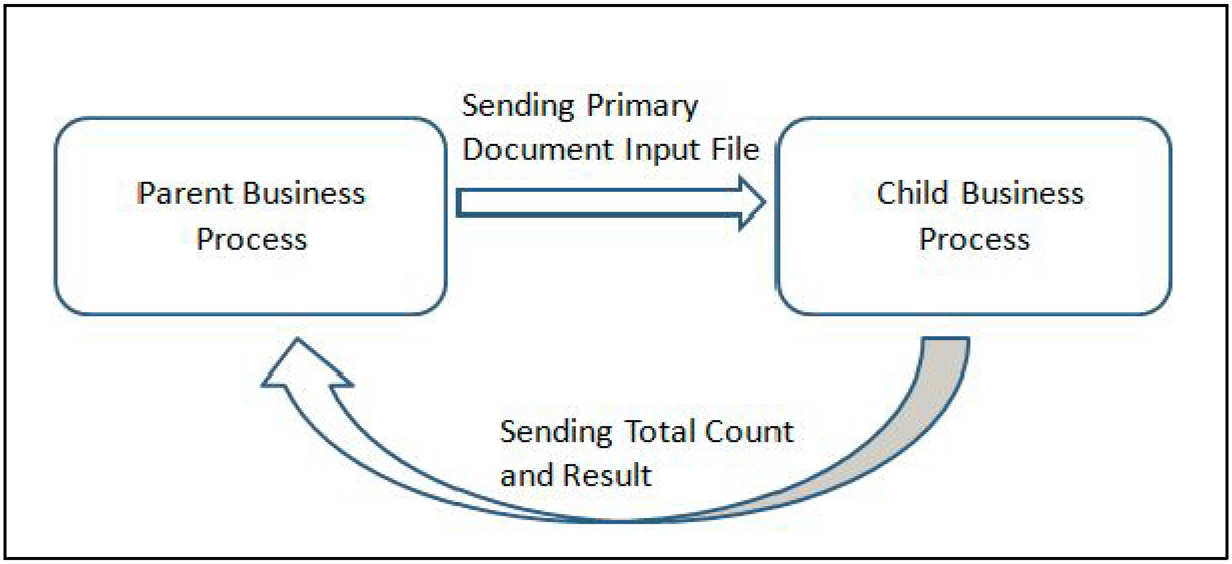
The Invoke Sub-Process service or Invoke Business Process in sync mode, the special tag that is called ‘message_to_child/message_to_parent' enables to pass along only the specified process data of the parent process or child subprocess. Using this tag can provide significant performance improvement.

Pass message from parent to child, to invoke a subprocess, create a special tag that is called ‘message_to_child' in the parent process, and append the required process data to the subprocess under this node. The Invoke Sub-Process service passes only this node to the subprocess. Similarly, the required message from the child can be passed to parent by using the ‘message_to_parent’ tag.

Exercise 2.3.4: Using message tags

Introduction

Consider a scenario where the parent business process needs to pass the inventory details as an input to the child process 'Basicinventoryprocess' to execute and the child subprocess needs to send back only the total count and the result to the parent business process. The loop back information between parent and child process.



For ease of understanding, the scenario is divided into four parts as listed:

- Create a parent business process
- Create a child business process
- Check in the parent and child business process
- Execute the parent business process and viewing the results

Note

For this exercise, use the existing business process from the C:\6F89G\Labfiles and modify according to this scenario.

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Exercise 2.3.4:Using message tags

(Continued)

Instructions

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For ease of understanding, the scenario is divided into four parts as listed:

- Create a parent business process
- Create a child business process
- Check in the parent and child business process
- Execute the parent business process and viewing the results

Note

For this exercise, use the existing business process in the Lab Files and modify according to this scenario.

Creating a Parent Business Process.

Step 1: Select **Business Processes** > **Manager** from the **Administration Menu**. The **Business Process Manager** page is displayed.

Step 2: Click **Go!** under Graphical Modeling.

Step 3: Enter User ID as **admin** and Password as password. The Graphical Process Modeler opens.

Step 4: Browse and open **AT_sync** business process.

Step 5: Drag and drop **Assign** activity from the BPML activities list and connect between XML encoder and Invoke Business Process Service and shown in the following diagram.

Step 6: Click **Assign** activity. The **Property Editor** for assign activity is displayed.

Step 7: Enter the parameters from the following table in the Property Editor.

Field	Value
From	//PrimaryDocument
To	message_to_child

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Exercise 2.3.4:Using message tags

(Continued)

Instructions

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Step 8: Click Invoke Business Process Service. The Service Editor for Invoke Business Process Service is displayed.

Step 9: Type WFD_NAME as **AT_Basicinventoryprocess1**.

Note

The process name AT_Basicinventoryprocess1 are checked in the following steps.

Step 10: Select **File** > **Save As**.

Step 11: Click **Yes** for Validate on save.

Step 12: Enter the File name as **AT_Message.**

Step 13: Save the business process in the **C:/Lab Files** folder. The business process is validated successfully.

Creating a Child Process

Step 1: Browse and open **AT_basicinventoryprocess** business process in the Graphical Process Modeler.

Step 2: Drag and drop **Assign** activity from the BPML activities list and connect between Assign and Sequence End and shown in the following diagram.

Step 3: Double-click **Assign** activity. The **Property Editor** for assign activity is displayed.

Step 4: Enter the parameters from the following table in the Property Editor.

Field	Value
From	//totalcount | //results
To	message_to_parent

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Exercise 2.3.4:Using message tags

(Continued)

Instructions

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Step 5: Select **File** > **Save As**.

Step 6: Click **Yes** for Validate on save.

Step 7: Enter the File name as **AT_Basicinventoryprocess1.**

Step 8: Save the business process in the **C:\6F89G\Labfiles** folder. The business process is validated successfully.

Note

You also directly pass the process data element name to the parent business process without using message_to_parent tag.

Check in the Parent and Child Business Process

Step 1: Select **Business Processes** > **Manager** from the **Administration Menu**, The Business Process Manager page is displayed.

Step 2: Click **Go!** next to Process creation, under **Create**.

Step 3: In the Process Name page, type **AT_Basicinventoryprocess1**.

Step 4: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 5: Browse to the **C:\6F89G\Labfiles** directory and select **AT_Basicinventoryprocess1.bp** for the file name.

Step 6: Enter **Child business process** for the description and select **Next** to continue.

Step 7: Accept the defaults. Click **Next** to continue.

Step 8: Do not set a deadline for this process, click **Next** to continue.

Step 9: Accept the default for life span and click **Next** to continue.

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Exercise 2.3.4:Using message tags

(Continued)

Instructions

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Step 10: Verify that the business process is enabled and click **Finish**.

Step 11: Click **Return**. The **Business Process Manager** page is displayed.

Step 12: Click **Go!** next to Process creation, under **Create.**

Step 13: In the Process Name page, type **AT_Message**.

Step 14: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 15: Browse to the **C:\6F89G\Labfiles**directory and select **AT_Message.bp** for the file name.

Step 16: Enter **Parent business process** for the description and select **Next** to continue.

Step 17: Set the persistence level to be **Full** and accept the remaining defaults, click **Next** to continue.

Step 18: Do not set a deadline for this process, click **Next** to continue.

Step 19: Accept the default for life span and click **Next** to continue.

Step 20: Verify that the business process is enabled and click **Finish**.

Step 21: Click **Return**. The **Business Process Manager** page is displayed.

Execute and View Results

Step 1: In the Business Process Manager page, type **AT_Message** and click **Go!** next to **Process Name**, under Search.

Step 2: Click the **execution manager** for the business process.

Step 3: Click **execute**.

Step 4: In the Filename dialog box, browse to the folder **C:\6F89G\Labfiles** and select the file **InventoryOf11.xml** and then click **Go!**. Observe the separate Execute Business Process window displayed while the business process is running.

Step 5: Close the Execute Business Process window when the business process is complete.

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Exercise 2.3.4:Using message tags

(Continued)

Instructions

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Step 6: From Sterling B2B Integrator Administration Menu, select **Business Process > Current Processes**.

Step 7: Click the ID for AT_Message business process. The **Business Process Detail** page is displayed.

Step 8: View the **Instance Data** for each Service in the AT_Message business process.

Step 9: Click the Subprocess ID. The **Sub-Process Detail** page is displayed.

Step 10: View the **Instance Data** for each Service in the AT_Basicinventoryprocess1 sub process.

Note

Use **Message To Service** and **Message From Service** links at each Instance data page to view the data to other services.

Result

The Instance Data for the invoke_test in the **Sub-Process Detail** page is only the primary document, as the AT_Message business process (parent) is configured to pass only the primary document to the AT_Basicinventoryprocess business process (child).

The **Message From Service** of the Instance Data for the invoke_test in the **Business Process Detail** page displays the total count and result as configured in the AT_Basicinventoryprocess business process.

Thread Monitor

Introduction

Monitor threads that are related to business processes in Sterling B2B Integrator by using the Thread Monitor. The Thread Monitor lists all threads currently running in the Activity engine. This information is useful for troubleshooting business processes that are stopped or hung. You can also stop a thread from the Thread Monitor.

Note

Whenever possible, stop business processes by using the options in the Business Process Monitor. These options allow the processes to stop gracefully, maintaining state and status information.

Monitor business process threads

To monitor threads, from the Administration Menu, select **Operations > Thread Monitor**. Sterling B2B Integrator shows the following information for each thread:

- The state of the thread
- The ID of the thread (click the ID number for more information about the thread.)
- The type of the thread
  - Business Process – click the ID number in the ID column to view the Business Process Detail page.
  - Remote Management Interface (RMI) – click the ID number in the ID column to view the Business Process Detail page.
  - Harness – click the ID number in the ID column to view the Business Process Detail page.
  - Schedule – click the ID number in the ID column to view the Schedule settings.
- The processing priority that is assigned to the thread.
- The date and time the thread was registered.

Note

Whenever possible, stop business processes by using the options in the Business Process Monitor. These options allow the processes to stop gracefully, maintaining state and status information.

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Thread Monitor

(Continued)

Stop threads

- To stop a thread, click the stop icon for the thread. This action instructs the thread to immediately stop processing without waiting for any steps to complete or exit with an error. In most cases, the thread stops immediately. The thread is eventually marked as interrupted.
- In some cases, it is not possible to stop a thread and your attempt is unsuccessful. Stopping a thread does not stop the service. The service is still available for Sterling B2B Integrator to use in other business processes.

Important

Terminate a thread as a last resort.Contact IBM Customer Support for assistance. Unexpected results can occur that can affect other processing. Stop or terminate business processes from the Business Process Monitor whenever possible.

Refresh option for thread monitor

By default, the Thread Monitor is set to automatically refresh every 15 seconds. To turn off this option, clear the check box next to Automatically refresh every 15 seconds.

Exercise 2.3.5: Sleep service

Introduction

In this exercise, Angela configures sleep service with a 180-second delay in continuing to the next step in the process.

Instructions

Complete the following steps to check in and execute business process with sleep service:

Step 1: Select **Business Processes** > **Manager** from the **Administration** Menu. The **Business Process Manager** page is displayed.

Step 2: Click **Go!** next to Process Definition under Create.

Step 3: Type **AT_Sleep** as **Name**.

Step 4: Select Check in Business Process created by the graphical modeling tool and click **Next**.

Step 5: Go to the **Lab Files** folder and select **AT_sleep.bp** for the file name.

Step 6: Enter **sleep service** for the description and click **Next** to continue.

Step 7: Set the persistence level to be **Full** and take the remaining defaults, click **Next** to continue.

Step 8: Do not set a deadline for this process, click **Next** to continue.

Step 9: Take the default for life span, click **Next** to continue.

Step 10: Verify that the business process is enabled and click **Finish**.

Step 11: Click **Return**. The Business Process Manager screen is shown.

Step 12: In the Search text box, type the **AT_Sleep** and click **Go!**.

Step 13: Click the **execution manager** icon for the process.

Step 14: Click the **execute** icon.

Step 15: Click Browse and go to the location of the test data file at **C:\6F89G\Labfiles\InventoryOf11.xml**. Observe the separate Execute Business Process window displayed while the business process is running.

Step 16: Close the Execute Business Process window when the business process is complete.

Modeling Guidelines for Large File Handling

(Continued)

Processing large files

To ensure efficient processing of large files, you must configure your business process models to prevent unnecessary overloading of the database, which can lead to production down scenarios. It provides information about business process modeling for scenarios that involve large file handling.

Configure Document Loops with Assign to message_to_child

Using the For Each Document service along with an Invoke BP service in a loop can be a source of production down issues for files of any size, but particularly for large files.

A considerable number of documents that are repeating through the For Each Document service, followed by an Invoke Business Process service, which carries forward all parent documents and process data, causes systems to become overwhelmed by data volume that grows on each cycle through the loop.

Each child process has a copy of the parent process' original document, and the N documents to cycle through the loop, N times the number of documents carry forward. For large documents or many documents, can fill the database.

To avoid this problem, structure your business process model as indicated in one of the two methods described here:

Method 1

In your business process model, include:

- An Assign activity with a message_to_child element that selectively passes the correct document (and any other selected data) to the next cycle in the loop
- A Release Service placed to clear out unneeded document copies

Note

Do not place the Release service between the Assign activity to message_to_child and the Invoke BP service. It removes the desirable data.

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Modeling Guidelines for Large File Handling

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Method 2

In your business process model, include one or more assign elements to message_to_child in the Message To Service of the Invoke Business Process service, to specify the correct document as an output element. You do not need to use the Release service if you use this configuration.

Note

The Sterling B2B Integrator BPML validation checks on business process models report an error for improper document loop configuration. However, if you are using business process models that are configured with this document loop that pre-date the validation check, and you checked out the process models and back in again, you can continue to run them. However, recommended practice is to modify their structure to ensure optimum processing based on the instructions.

Lesson review

What you have been able to do

After completing this lesson, you should have been able to:

- Describe basic business process concepts.
- Monitor and manage business processes.
- Set performance-related options on the check-in screen.
- Manually run a business process and track results.
- List guidelines for business process design and efficient processing.
- Describe the thread monitor function.
- Explain the guidelines for large file handling.